

Introduction

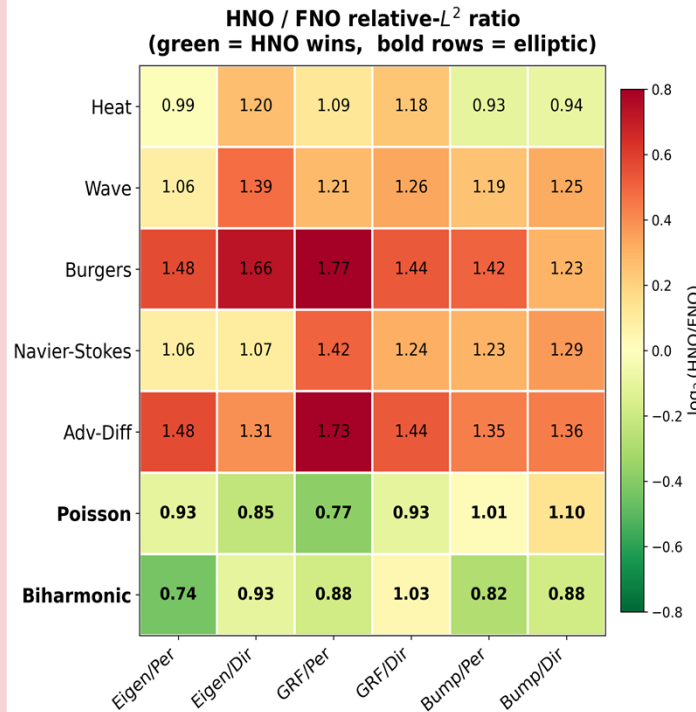
- Neural operators learn PDE solution operators; the Fourier Neural Operator (FNO) works in the complex Fourier basis.
- For real-valued PDE solutions the complex FFT is redundant — conjugate symmetry fixes half the spectrum.
- We introduce the Hartley Neural Operator (HNO): the exact real-valued mirror of FNO, built on the Discrete Hartley Transform.
- Thesis:** the best spectral basis is a property of the operator, not a universal choice.

Methods

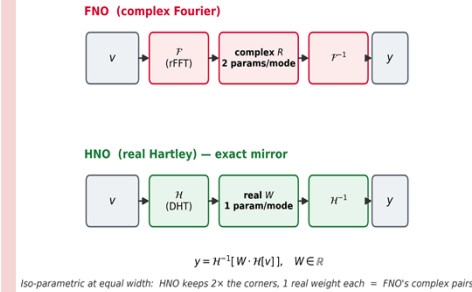
- Clean real-diagonal HNO: one real weight per retained mode, no complex arithmetic — $y = H^{-1}[W \cdot H[v]]$, W real.
- Iso-parametric with FNO at equal width: HNO keeps 2× the spectral corners but one real weight where FNO carries a complex pair.
- Matched training (identical optimizer, learning rate, schedule, gradient clip, width) — any gap is the basis, not tuning.
- Benchmark: 7 PDEs × 3 initial-condition families × periodic / Dirichlet BCs, 64² grid, relative-L², single A100.

Results

HNO / FNO relative-L² across the full benchmark. Green = HNO wins; bold rows = elliptic. Rows are elliptic.



HNO is the exact real mirror of FNO



Key findings

- Elliptic → HNO.** Real symmetric Green's functions are diagonalized exactly by one real weight per mode.
- Time-dependent → FNO.** Oscillation and transport carry phase a real diagonal cannot represent.
- Monotone in phase.** Biharmonic = largest HNO win; Burgers / Adv-Diff = largest FNO win; heat ≈ tie.
- IC sets magnitude, not sign.** The basis is a property of the operator, not the data.

The split falls exactly where the theory predicts.

Self-adjoint elliptic operators have real, symmetric Green's functions that the real Hartley multiplier diagonalizes exactly. Time evolution adds phase — oscillation, transport — that a real diagonal cannot represent.

Best-seed shown; preliminary 64² grid, single representative parameter.

Theory: why the winning basis tracks operator symmetry

Elliptic ⇒ HNO is exact

$$\hat{u}(k) = \hat{G}(k) \hat{f}(k), \quad \hat{G}(k) \in \mathbb{R}$$

self-adjoint L : the Green's function is real & symmetric, so a real multiplier $W(k)$ reproduces $\hat{G}(k)$ with no error.

Phase ⇒ FNO is needed

$$\hat{G}(k) = |\hat{G}(k)| e^{i\theta(k)}, \quad \theta(k) \neq 0$$

transport / oscillation give a nonzero phase $\theta(k)$ that a purely real diagonal multiplier structurally cannot encode.

Monotone ordering

$$\Delta(\text{HNO}, \text{FNO}) \propto \langle |\theta(k)| \rangle$$

the HNO-FNO gap grows with the operator's phase content ⇒ predicts the elliptic ↔ time-dependent split.

Conclusions

- Match the spectral basis to the symmetry of the solution operator.
- Real Hartley basis — self-adjoint, phaseless problems (elliptic solves, diffusion).
- Complex Fourier basis — oscillatory or advective problems.
- HNO matches FNO's cost and parameter count with no complex arithmetic.
- Future: native fast Hartley transform; basis from detected operator symmetry; non-spectral & DeepONet baselines.

References

- Li et al. Fourier Neural Operator for Parametric PDEs. ICLR 2021.
- Lu et al. Learning nonlinear operators (DeepONet). Nat. Mach. Intell. 2021.
- Bracewell. The Discrete Hartley Transform. J. Opt. Soc. Am. 1983.
- Wong et al. HartleyMHA. MICCAI 2023.
- Kovachki et al. Neural Operators. JMLR 2023.

Code Repository

github.com/jaysulk/hartley-neural-operator

